$$= |z_1|^2 + 2\,\mathrm{Re}(z_1\overline{z}_2) + |z_2|^2 \qquad (\because\ 2\,\mathrm{Re}(z) = z + \overline{z})$$

$$\leq |z_1|^2 + 2|z_1\overline{z_2}| + |z_2|^2 \qquad (\because\ \mathrm{Re}(z) \leq |z|)$$

$$= |z_1|^2 + 2|z_1|\,|z_2| + |z_2|^2 \qquad (\because |z_1 z_2| = |z_1|\,|z_2| \text{ and } |z| = |\overline{z}|)$$

$$\Rightarrow\ |z_1 + z_2|^2 \leq \left(|z_1| + |z_2|\right)^2$$

$$\Rightarrow |z_1 + z_2| \leq |z_1| + |z_2|.$$

Geometrical interpretation

Now consider the triangle shown in figure with vertices O, z_1 or z_2, and $z_1 + z_2$. We know from geometry that the length of the side of the triangle corresponding to the vector $z_1 + z_2$ cannot be greater than the sum of the lengths of the remaining two sides. This is the reason for calling the property as "Triangle Inequality".

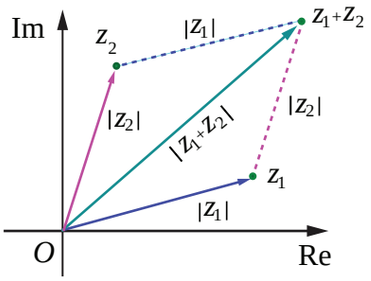


Fig. 2.17

It can be generalized by means of mathematical induction to finite number of terms:

$|z_1 + z_2 + z_3 + \cdots + z_n| \leq |z_1| + |z_2| + |z_3| + \cdots + |z_n|$ for $n = 2, 3, \cdots$.

Property The distance between the two points z_1 and z_2 in complex plane is $|z_1 - z_2|$

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$|z_1 - z_2| = |(x_1 - x_2) + (y_1 - y_2)i|$$

$$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\,.$$

Remark

The distance between the two points z_1 and z_2 in complex plane is $|z_1 - z_2|$.

If we consider origin, z_1 and z_2 as vertices of a triangle, by the similar argument we have

$|z_1 - z_2| \leq |z_1| + |z_2|$

$\big||z_1| - |z_2|\big| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ and

$\big||z_1| - |z_2|\big| \leq |z_1 - z_2| \leq |z_1| + |z_2|$.

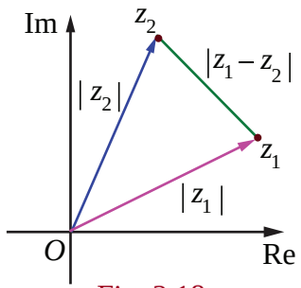


Fig. 2.18

Property Modulus of the product is equal to product of the moduli.

For any two complex numbers z_1 and z_2, we have $|z_1 z_2| = |z_1||z_2|$.

Proof

We have $|z_1 z_2|^2 = (z_1 z_2)(\overline{z_1 z_2})$ $\qquad (\because |z|^2 = z\,\overline{z})$

$= (z_1)(z_2)(\overline{z_1})(\overline{z_2})$ $\qquad (\because \overline{z_1 z_2} = \overline{z_1}\,\overline{z_2})$